

The reliability plane

Complete product requirements for turning MCM inside out: prove the telemetry, let customers declare placement, failover, release, and SLO as policy, put agents on the pager with a warrant, and sell the fourth nine. Companion to the Strategy and the due-diligence brief.

NORTH STAR

Every production workload on Baseten runs under a declared, observable policy — measured as **MTTR on agent-covered incidents, goodput at declared SLO, policy adoption across dedicated accounts, and SLA-ladder revenue**.

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REFERENCE IMPLEMENTATION:
[GITHUB.COM/VSIWACH/BASETEN-RELIABILITY-
PLANE](https://github.com/vsiwach/baseten-reliability-plane)

EVERY FEATURE NAMES ITS
STRATEGY
PILLAR (P1-P6) AND MVP HOOK

01 Product definition

Product: the reliability plane — four declared policy objects (Placement, Failover, Release, SLO) enforced by MCM and the router, observed in a single operate console, operated by agents under a closed allowlist, and sold as an SLA ladder (Managed 99.9 / Declared 99.95 / Operated 99.99). **Available to two buyers from day one:** teams evaluating Baseten for dedicated inference (the console is their evaluation harness — agentic onboarding, real SLO baselines, certified migration when the evidence says so) and existing customers running Baseten inside a multicloud portfolio (the console is their daily reliability surface, and the ledger their case for consolidating traffic).

The trust machinery is the observability contract: machine-readable failure reasons, capacity-wait states, metrics completeness timestamps, and published rate-limit budgets — the console never lies, so the policies can be believed.

The retention machinery is accumulated policy: declared SLOs, placement rules, release gates, and incident history compound switching costs the way raw endpoints never do.

The adoption machinery is defaults-with-override: every policy object ships pre-filled with what Baseten's operators would do anyway; declaring is one edit, not a config project.

The acquisition machinery is the migration play: read-only monitoring of the clouds customers already run on (the wedge nobody is suspicious of), a route ledger that accumulates cross-provider SLO and cost evidence, and certified migration offered exactly when the evidence — and the customer — are ready. Control features stay Baseten-side: observe → migrate → declare → operate.

Supporting metrics: % dedicated workloads under ≥1 declared policy; operate-console WAU; MTTR p50/p95 on agent-covered incidents; goodput at declared SLO; zero-downtime deploys as % of promotions; preemption events honored per policy; external pools attached and traffic won back on measured evidence; SLA-ladder ACV.

02 Users

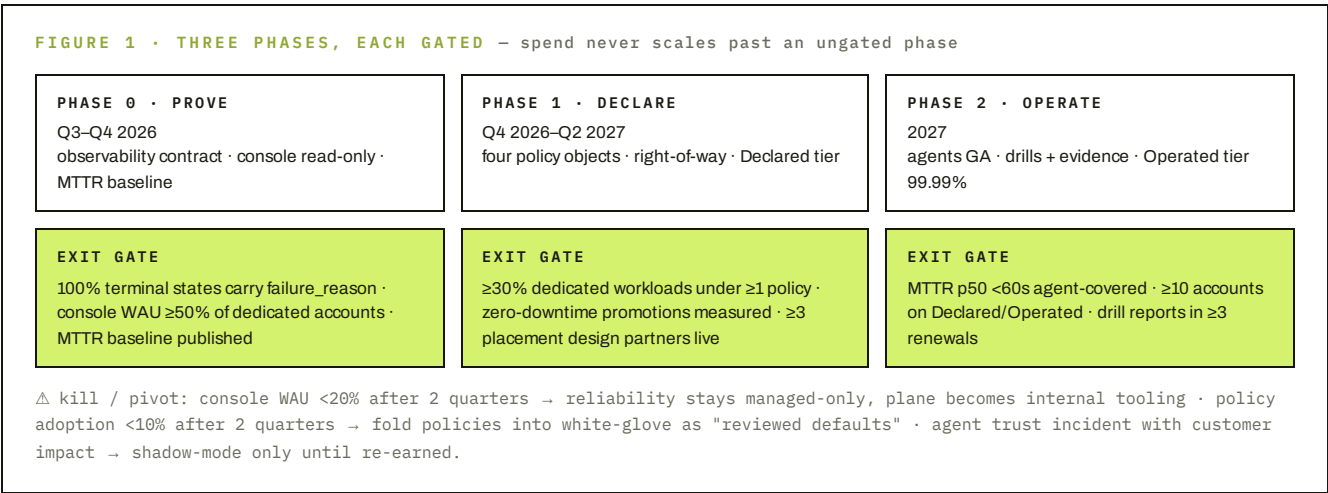
PERSONA	JOB TO BE DONE	CONVERTS VIA
P1 · AI-native inference owner PRIMARY	Tokens are COGS; downtime is churn. Needs SLOs held during traffic spikes and model rollouts, and proof that reliability didn't regress after every change.	Console → SLO + Release policy → Declared tier
P2 · Enterprise / healthcare platform owner PRIMARY	Needs residency and compliance provable to procurement and auditors; needs failover that is policy, not anecdote; needs evidence artifacts, not assurances.	PlacementPolicy + drills → Operated tier
P3 · On-call engineer	Needs to tell slow from stuck from doomed in one glance; needs safe remediations (replace replica, shift region, rollback) without a support ticket at 3am.	Incident console → agent remediation
P4 · The agent NEW	Machine consumer of every surface: needs failure reasons, completeness timestamps, Retry-After, and idempotent actions — the same contract gaps that break dashboards break agents first.	MCP server + allowlisted action APIs
P5 · Baseten's own operators	Today's white-glove work (encoding constraints, watching failovers) becomes reviewing declared policy and auditing agent actions — the scaling lever for the ops team itself.	Internal adoption gates GA

Goals and non-goals

Goals (1) The observability contract closed and versioned; (2) the operate console as the daily surface for dedicated accounts; (3) all four policy objects GA with defaults-with-override; (4) right-of-way for compliance-bound workloads live on sensitive capacity; (5) measurable MTTR decline via agent remediation; (6) the SLA ladder monetizing declared reliability.

Non-goals Engine work (BIS-LLM performance is its own roadmap); multi-cloud abstraction resale; a generic APM product (export to Datadog stays first-class); agent autonomy beyond the closed allowlist; per-token pricing changes; replacing MCM's scheduler — the plane declares intent, MCM keeps enforcing it.

03 The roadmap and its gates



04 Phase 0 — Prove Q3–Q4 2026 · P1

Entry: now. **Exit gate:** failure reasons on 100% of terminal states + capacity-wait states live + metrics completeness shipped + console WAU ≥50% of dedicated accounts + MTTR baseline published. **Scope note:** every item below is a first-hand-verified gap (friction log #, in evidence); Phase 0 is deliberately unglamorous — it is the trust the rest of the plane stands on, and it is the migration play's front door: an evaluator who hits friction #6 or #7 in their first hour is a migration lost before any evidence could speak.

FEATURE	REQUIREMENT	TARGET METRIC	MVP HOOK
F0.1 Failure-reason contract	Every deployment terminal state carries a machine-readable failure_reason (BUILD_FAILED cause, OOM, dep-skew, SKU-gated); build logs via management API; push-time pre-flight (SKU allowed? weights fit host RAM? python/quant combo valid?). friction #7 #12 #13 #14	0 terminal states without reason; pre-flight catches ≥80% of the failure classes seen in the log	status API in sim; failure taxonomy in events.json
F0.2 Capacity-wait states	WAITING_FOR_CAPACITY as first-class status with queue context and alternative-SKU suggestion; a deploy never dies INACTIVE without a reason; wait time visible in console and API. friction #6	0 silent INACTIVE transitions; median time-to-diagnosis for placement waits <1 min	placement feed panel; capacity events in replay
F0.3 Metrics you can build on	data_complete_through timestamp on every metrics response (silent nulls become labeled lag); published rate-limit budgets sized for dashboards; Retry-After + X-RateLimit-* on every 429 platform-wide; token metrics for all deployments, not just BIS-LLM. friction #10 #18 #19	0 unlabeled empty windows; a 60s-interval console polls a 20-deployment workspace with headroom	console's "no data YET vs no traffic" rule; honest-numbers banner

FEATURE	REQUIREMENT	TARGET METRIC	MVP HOOK
F0.4 Operate console (read-only) + bring-your-endpoint	One screen per workspace: SLO hero (goodput, p99 TTFT/TPOT, \$/Mtok measured), pool grid, placement feed with reasons, release state, incident timeline. Every number measured, never estimated; every silence labeled. Bring-your-endpoint: any OpenAI-compatible deployment on any provider attaches read-only as a monitored pool — the console is useful before a single request moves to Baseten, which is the point (the wedge for F2.5). Monitor-only is a hard product rule: no control affordance ever renders on an external deployment.	WAU ≥50% of dedicated accounts; ≥25% of console accounts attach ≥1 external endpoint; <5s to answer "is my model healthy and why"	<code>operate.html</code> – the reference implementation IS the design contract; <code>modal pool = monitor-only exemplar</code>
F0.5 Alerting primitives	SLO objects with burn-rate alerts; webhook + PagerDuty/Slack integration; deploy-event notifications. Absence today is verified — customers export to their own stack or fly blind.	≥40% of dedicated accounts with ≥1 alert rule	<code>slo-policy.yaml</code> thresholds drive console badges
F0.6 Agentic onboarding — on Baseten's own rails	One prompt to the customer's coding agent (Claude Code, Cursor, any MCP client) onboards them: connect the Baseten MCP server (workspace, autoscaling, logs — shipped June 30 2026) and Baseten agent skills (deploy, Truss, tune), attach external endpoints read-only, baseline every route into the ledger, propose the first A/B candidate — stopping before any traffic moves. Credit where due: Baseten made the platform agent-operable first; this feature is the onboarding those rails deserve, and the on-ramp for F2.5's migration play.	Time from prompt to attached + baselined workspace <10 min; ≥50% of new plane accounts onboard via agent; 0 write actions before human confirmation	<code>onboard.html</code> – runnable prompt + 4-step flow + credit block

05 Phase 1 — Declare Q4 2026–Q2 2027 · P2 P3 P4

Entry: Phase 0 exit. **Exit gate:** ≥30% of dedicated workloads under ≥1 declared policy; zero-downtime promotions measured as default; ≥3 placement design partners (≥1 healthcare) live; Declared tier launched. **Kill/pivot:** policy adoption <10% after 2 quarters → policies become operator-reviewed defaults, not customer surface.

FEATURE	REQUIREMENT	TARGET METRIC	MVP HOOK
F1.1 SLOPolicy P0 · THE KEYSTONE	Declare p99 TTFT/TPOT, error rate, and goodput targets per deployment/route. SLOs drive autoscaling signals (scale to the SLO, not just concurrency), alert thresholds, release gates, and agent triggers — one declaration, four consumers.	≥50% of declared workloads meet SLO ≥99% of windows; SLO-driven scaling beats concurrency-only on p99 in A/B	slo-policy.yaml consumed by hero, gates, and agent in replay
F1.2 ReleasePolicy P0	Standing traffic splits beyond promotion: canary steps with metric gates and auto-rollback; shadow/mirror with diff reports (zero user risk); A/B with cohort stickiness; declarative warm-up spec and drain semantics; probe presets per engine replacing 30-min defaults; Chains parity. today: promotion-scoped, default-off, no Chains	99.9%+ of promotions drop zero requests; auto-rollback <5 min from gate breach; ≥1 shadow eval per major model migration	release-policy.yaml + release track panel; canary steps animate in replay
F1.3 PlacementPolicy P0 · THE HEADLINE	One object: allowed regions, compliance regime tags (HIPAA, EU-residency), provider preference, capacity tier order (reserved → on-demand → spill). Self-serve via API/UI/truss config; enforced by MCM; every placement decision emitted with a reason to the placement feed. Replaces "manually encoded per customer."	≥3 design partners; 0 placement violations; regional-env setup time from support-ticket days to <1 hour self-serve	placement-policy.yaml + placement feed with reasons
F1.4 Right-of-way / priority classes P1	Priority classes per workload; compliance-bound workloads preempt region-flexible ones on sensitive capacity; preempted workloads drain gracefully and relocate per their own policy; preemption events visible to both parties.	Sensitive-capacity availability for compliance-bound ≥99.9% during contention drills; 0 SLO breaches caused by preemption on relocated workloads	priority field + contention drill in replay
F1.5 FailoverPolicy P1	Declared topologies: active-active regions, ordered fallback, spill targets (second deployment or Model APIs for open-weight routes); health-aware recovery with fast probe defaults; failover events on the incident timeline. today: MCM behavior, invisible	Failover RTO <60s in drills; ≥1 customer-visible failover evidence report per event	failover-policy.yaml + "kill a pool" drill button
F1.6 Warm capacity as policy P2	Scheduled and forecast-driven pre-warm replacing cron-the-API guidance; BDN weight-caching default-on for hf://-resolvable models (measured 2.4× cold-start cut); billing labels cold-start vs serving minutes. friction #17 + docs' own cron advice	p95 cold start for cached 8B-class models <60s; 0 accounts pre-warming via cron	cold-start economics in pool cards

06 Phase 2 — Operate 2027 · P5 P6

Entry: Phase 1 exit + three design partners on declared policy for a full quarter. **Exit gate:** MTTR p50 <60s on agent-covered incident classes; ≥10 accounts on Declared/Operated tiers; drill evidence used in ≥3 renewals. **Kill/pivot:** any agent action causing customer-visible harm → agents to shadow mode (recommend-only) until a full quarter clean.

FEATURE	REQUIREMENT	TARGET METRIC	MVP HOOK
F2.1 Incident console + agent remediation P0 · THE HEADLINE	Detection from declared SLOs (never just /healthz — a pool can pass health checks while serving garbage); agents act under a closed allowlist (quarantine, probe, reinstate, scale, shift, rollback, escalate) with hard guards (never orphan the last healthy pool; sticky quarantine; escalate once, then slow-poll); every action on a replayable timeline; MTTR detect → diagnose → resolve tracked per incident. Shadow mode (recommend-only) is the GA on-ramp.	MTTR p50 <60s agent-covered vs baseline; 0 guard violations; ≥70% of eligible incidents resolved without human action	the 8.8–9.2s measured loop, replayed from committed episodes JSONL
F2.2 Drills + evidence P1	Scheduled chaos drills (kill pool, degrade region, exhaust capacity) run against declared policies; per-drill and per-incident signed evidence reports (what happened, what policy required, what the plane did, MTTR) — an artifact the customer's auditor accepts.	≥1 drill/quarter per Operated account; evidence artifact attached to 100% of incidents	drill button → evidence card in console
F2.3 The SLA ladder P0 · MONETIZATION	Managed 99.9 (today) / Declared 99.95 (policy + reserved floor, capacity terms inside the SLA) / Operated 99.99 (multi-region + agents + drills). The 99.99% marketing already claims becomes purchasable with evidence as the receipt.	≥10 accounts on paid tiers; 0 SLA misses on Operated; ladder ACV growth	tier badge in console header
F2.4 Cost governance P1	Measured \$/Mtok per deployment (pool \$/hr ÷ measured tok/s — never estimated); per-policy and per-team attribution; budget caps with policy-aware enforcement (degrade to cheaper pool before throttling).	100% dedicated deployments with measured \$/Mtok; ≥50% of enterprise accounts using chargeback	/v1/costs pattern; \$/Mtok in hero + pool cards
F2.5 Route ledger + certified win-back P1 · THE MIGRATION PLAY	The ledger accumulates from F0.4 monitoring: per-route workload shape, SLO adherence, measured \$/Mtok per provider. When a monitored external route would hold its SLO at lower measured \$/Mtok on Baseten capacity, the console offers certified migration — shadow → certify → promote, reversible for the route's life. The trust ladder is explicit: observe (F0.4) → migrate (this) → declare (F1.x) → operate (F2.1) — control-plane features apply only to Baseten-side workloads and are never the day-one ask. Guardrails: cards only from rerunnable measured evidence; the customer owns and can export their ledger; external legs at zero markup; monitor-only means monitor-only.	≥25% of accounts with external endpoints attached (F0.4); ≥40% of win-back cards accepted into shadow; measured traffic migrated per quarter; 0 disputed recommendations	win-back card on migration panel with measured delta; monitor-only badge on modal pool

07 Always-on tracks

Honest numbers. Every metric on any surface is measured, never estimated; empty windows are labeled lag or no-traffic, never faked; savings are reported against declared baselines, never invoiced from counterfactuals. This rule is inherited by the reference implementation and is non-negotiable — a reliability product that fabricates a number once is dead.

Agent governance. The allowlist is versioned and auditable; learning tunes cadence and thresholds, never expands authority; every agent action is replayable end-to-end; a public MTTR benchmark methodology invites external rerun.

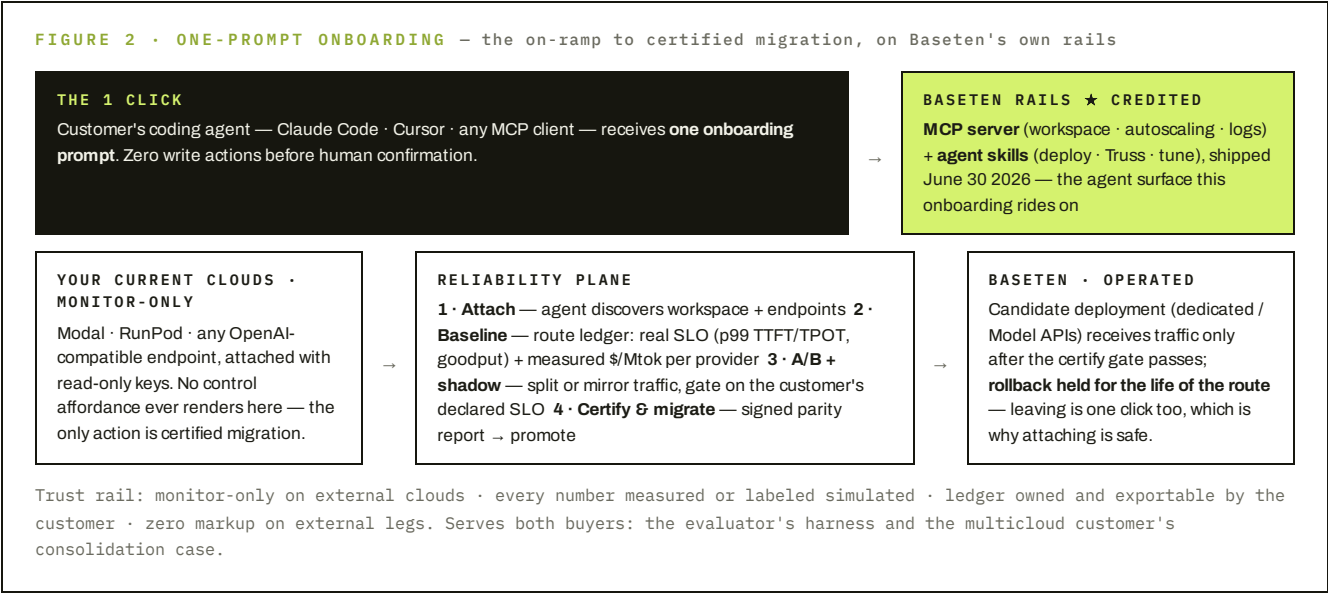
Contract stability. The observability contract (F0.1–F0.3) is versioned API surface with deprecation policy — agents and consoles are customers too (persona P4), and breaking them silently is a reliability incident.

08 Success metrics by phase

PHASE 0 · Q3–Q4'26	PHASE 1 · Q4'26–Q2'27	PHASE 2 · 2027
100% terminal states with failure_reason; 0 silent INACTIVE; metrics completeness shipped; console WAU ≥50% of dedicated accounts; MTTR baseline published; ≥40% accounts with alert rules.	≥30% dedicated workloads under ≥1 policy; zero-downtime promotions default; ≥3 placement partners (≥1 healthcare); failover RTO <60s in drills; 0 cron pre-warming; Declared tier live.	MTTR p50 <60s agent-covered; ≥70% eligible incidents auto-resolved; ≥10 SLA-ladder accounts; 0 Operated SLA misses; evidence in ≥3 renewals; public MTTR benchmark published.

09 The MVP: the reference implementation

Phase 0 defines the product surface; the reference implementation makes it tangible before a single platform engineer is committed. It is a static, Vercel-deployable operate console that replays *real recorded evidence* — the measured MTTR episodes, the chaos drills, the friction-log failure taxonomy — against the four policy YAMLs, built from the ai-native-pipeline base (release engine, incident agent, measured-cost ledger, policy schemas). Its architecture is the story in one picture:



Five commitments:

- 1 · One screen (F0.4):** SLO hero, pool grid, placement feed with reasons, release track, incident timeline — the operate.html design contract, glanceable in 5 seconds.
- 2 · Four policies as YAML (F1.1–F1.5):** slo-, release-, placement-, failover-policy.yaml with defaults-with-override semantics, viewable and editable in the console with simulated enforcement.
- 3 · The agent loop replayed (F2.1):** committed incident episodes drive the timeline — quarantine → probe → reinstate with the MTTR stopwatch; a drill button re-runs the loop live in simulation.

4 · Honest numbers throughout: every figure traces to a committed CSV/JSONL or is labeled SIMULATED; the silent-null rule is implemented, not just specified.

5 · Shareable by default: clean public repo, MIT, README mapping every panel to the JD's four outcome areas and to F-numbers in this PRD.

10 Open questions

MCM's policy readiness — how much of the scheduler already keys on per-customer constraints vs needs refactoring to honor declared intent? Phase 0 discovery with the MCM team decides Phase 1 sequencing.

Preemption economics — what compensation (credits, priority refund) makes right-of-way preemption acceptable to the preempted party without gaming?

Chains parity cost — rolling deployments are unsupported for Chains today; is that architectural or just unbuilt? It gates ReleasePolicy's "no excluded products" claim.

SLA actuarial basis — how many quarters of drill evidence does finance/legal need before selling 99.99% with capacity terms included, and at what reserved-floor threshold?

Agent liability — what does the Operated-tier contract say about an agent action that worsens an incident, and does shadow-mode history change the answer?

Enterprise-tier validation — the friction log is n=1 on a non-enterprise workspace; which walls (GPU gating, replica caps, metrics budgets) exist at enterprise tier? Phase 0 telemetry answers this with data.